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Subject : Cryptography for network analysis

CIA EXAM : 1

A=0, B=1, C=2, D=3
E=4, F=5, G=6, H=7
I=8, J=9, K=10
L=11, M=12, N=13
O=14, P=15, Q=16, R=17
S=18, T=19, U=20, V=21
X=22, W=23, Y=24
Z=25

1) Given:

Plaintext = "MEETAT NOON"

key: 7

From
A=0, B=1, ... Z=25

Cases cipher:-
The cypher text is used for the encryption & decryption using the given key value. From the alphabet the numeric value as been taken.

Encryption:-
Plaintext: M E E T A T NOO N
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
12 4 4 19 0 19 13 14 13

key: 7

Formula:-

$C = (P+k) \bmod 26$

$C = (\overset{M}{12}+7) \bmod 26$

$C = 19 \bmod 26$

$C = 19 \Rightarrow T$

$C = (E+k) \bmod 26$

$C = (4+7) \bmod 26$

$C = 11 \bmod 26$

$C = L \Rightarrow 11$

$C = (E+k) \bmod 26$

$C = (4+7) \bmod 26$

$C = 11 \bmod 26$

$C = L \Rightarrow 11$

$C = (T+k) \bmod 26$

$C = (19+7) \bmod 26$

$C = 26 \bmod 26$

$C = 0 \Rightarrow A$

$C = (A+k) \bmod 26$

$C = (0+7) \bmod 26$

$C = 7 \bmod 26$

$C = 7 \Rightarrow H$

$C = (T+k) \bmod 26$

$C = (19+7) \bmod 26$

$C = 26 \bmod 26$

$C = 0 \Rightarrow A$

N
$C=(N+K) \bmod 26$

$C=13+7 \bmod 26$

$C=20 \bmod 26$

$C=20 \rightarrow U$

$C=(O+K) \bmod 26$

$C=(14+7) \bmod 26$

$C=21 \bmod 26$

$C=21 \rightarrow V$

ciphertext: ↑ L L A HA UVVU

19 11 11 0 7 0 20 21 20 21

② Decryption using the caesar cipher text:-

FORMULA: $P=(C-K) \bmod 26$

$P=(T-K) \bmod 26$

$P=(19-7) \bmod 26$

$P=12 \bmod 26$

$P=M \Rightarrow 12$

$P=(L-K) \bmod 26$

$P=(11-7) \bmod 26$

$P=E \rightarrow 4$

$P=(L-K) \bmod 26$

$P=(11-7) \bmod 26$

$P=E \rightarrow 4$

$P=(A-K) \bmod 26$

$P=0-7 \bmod 26$

$P=19 \bmod 26$

$P=19 \rightarrow T$

$P=(H-K) \bmod 26$

$P=(7-0) \bmod 26$

$P=7 \bmod 26$

$P=A$

$P=(A-K) \bmod 26$

$P=(0-7) \bmod 26$

$P=$ 0 7

$P=(V-K) \bmod 26$

$P=20-7 \bmod 26$

$P=13 \bmod 26$

$P=N$

$P=(V-K) \bmod 26$

$P=(21-7) \bmod 26$

$P=14 \bmod 26$

$P=14 \rightarrow O$

$P=(V-K) \bmod 26$

$=20-7 \bmod 26$

$=13 \bmod 26$

$P=N$

Decryption process of above cipher text:

Plain text: MEET AT NOON

ii) Vigenere cipher text:

Formula for Encryption: $C = (P_i + K_i) \bmod 26$

Formula for Decryption: $P = (C - K_i) \bmod 26$

Given

Plain text: "MEET AT NOON"
12 4 4 19 0 19 13 14 14 13

key: "SUN"
18 20 13

Encryption :-

$C = P_i + K_i \bmod 26$
$= (12 + 18) \bmod 26$
$= 30 \bmod 26$
$= 4 \Rightarrow E$

$C = (E + U) \bmod 26$
$= (4 + 20) \bmod 26$
$= 24 \bmod 26$
$= 24 \Rightarrow Y$

$C = (E + N) \bmod 26$
$= (4 + 13) \bmod 26$
$= 17 \bmod 26$
$= 17 \Rightarrow R$

$C = (T + S) \bmod 26$
$= (19 + 18) \bmod 26$
$= 37 \bmod 26$
$= 11 \Rightarrow L$

$C = (A + U) \bmod 26$
$= (0 + 20) \bmod 26$
$= 20 \Rightarrow U$

$C = (T + N) \bmod 26$
~~$= (T + 13)$~~
$= (19 + 13) \bmod 26$

$C = 6 \Rightarrow G$

$C = (N + S) \bmod 26$
$C = (13 + 18) \bmod 26$
$C = 5 \Rightarrow F$

$C = (O + U) \bmod 26$
$C = (14 + 20) \bmod 26$
$C = 34 \bmod 26$
$C = 8 \Rightarrow I$

$C = (O + N) \bmod 26$
$C = (14 + 13) \bmod 26$
$C = 1 \Rightarrow B$

$C = (N + S) \bmod 26$
$C = (13 + 18) \bmod 26$
~~$C = 31 \bmod 26$~~
$C = 5 \Rightarrow F$

The cypher text is: EYRI UG FIBF
4 24 17 11 20 6 5
key: 3 U N
18 20 13
8 1 5

Decryption :-

P = (E - S) mod 26
P = (4 - 18) mod 26
P = -14 mod 26
P = 12 ⇒ M

P = (Y - U) mod 26
= (24 - 20) mod 26
= 4 ⇒ E

P = (N - R) mod 26
P = (13 - 17) mod 26
P = H ⇒ E

P = (L - S) mod 26
P = (11 - 18) mod 26
P = 19 ⇒ T

P = (U - U) mod 26
P = 0 mod 26
P = 0 ⇒ A

P = (G - N) mod 26
P = (6 - 13) mod 26
P = 19 ⇒ T

P = (F - S) mod 26
P = (5 - 18) mod 26
P = 13 ⇒ N

P = (I - U) mod 26
P = (8 - 20) mod 26
P = 14 ⇒ O

P = (B - N) mod 26
P = (1 - 13) mod 26
P = -12 mod 26
P = 14 ⇒ O

P = (S - S) mod 26
P = (5 + 18) mod 26
P = 15 ⇒ N

The plaintext is : MEET AT NOON

9) comparision of ceasar cipher & vigenere cipher

Ceasar cipher	vigenere cipher
It uses the same key	It uses the key which was rotation
The key value is numerical	The key value was converted from Alphabetic to numeric value
The process was easy	The process was slight difficult.

8)

2) Given :-

Plaintext : "MATH"

key :- $\begin{bmatrix} 3 & 2 \\ 9 & 7 \end{bmatrix}$

Formula for Encryption : $C = PK \bmod 26$

Formula for Decryption : $P = CK^{-1} \bmod 26$

Encryption :-

$MA = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$

$C = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 9 & 7 \end{bmatrix} \bmod 26$

$= \begin{bmatrix} 36 & 24 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} \bmod 26$

$= \begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26$

$C = \begin{bmatrix} 10 \\ 8 \end{bmatrix} \rightarrow KI$

$TH = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$

$= \begin{bmatrix} 36 & 24 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} \bmod 26$

$= \begin{bmatrix} 57+14 \\ 95+49 \end{bmatrix} \bmod 26$

$= \begin{bmatrix} 71 \\ 144 \end{bmatrix} \bmod 26$

$\begin{bmatrix} 19 \\ 14 \end{bmatrix} \bmod 26$

$= \begin{bmatrix} 19 \\ 14 \end{bmatrix} \rightarrow OU$

Encryption was done the cipher text is : KITO
10 8 19 14

Decryption:-

$P = CK^{-1} \bmod 26$

$K^{-1} = \frac{1}{|A|} \text{adj } A$

$= |A| = 21 - 10 \Rightarrow 11$

$\text{adj } A = \begin{vmatrix} 7 & -2 \\ -5 & 3 \end{vmatrix}$

$K^{-1} = 209$

$209 \bmod 26 = 1$

$209 \bmod 208 = 1$

$11^{-1} = 19$

$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$

$K^{-1} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$

$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

$KI \Rightarrow \begin{bmatrix} 10 \\ 8 \end{bmatrix}$

$P = CK^{-1} \bmod 26$

$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} \bmod 26$

$P = \begin{bmatrix} 142 \\ 180 \end{bmatrix} \bmod 26$

$P = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \Rightarrow MA$

$TO \Rightarrow \begin{bmatrix} 19 \\ 14 \end{bmatrix}$

$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} \bmod 26$

$P = \begin{bmatrix} 57 + 196 \\ 171 + 70 \end{bmatrix} \bmod 26$

$P = \begin{bmatrix} 253 \\ 241 \end{bmatrix} \bmod 26$

$P = \begin{bmatrix} 19 \\ 7 \end{bmatrix} \Rightarrow TH$

$P =$ "MATH"

A) Given :-

$P = 11 \quad q = 19 \quad e = 7 \quad M = 5$

$n = P \times q = 11 \times 19 = 209$

$\phi(n) = (P-1)(q-1)$

$\phi(n) = (10)(18) \Rightarrow 180$

Encryption formula :-

$C = M^e \bmod n$

Decryption formula :-

$M = C^d \bmod n$

To find the d

$1 < e < n$

$1 < 7 < 180$

$d = 103$

To find Public key $(e, n) \Rightarrow (7, 180)$

To find Private key $(d, n) \Rightarrow (103, 180)$

gcd = (7, 25)

= 5

5×5 = 25

25×5 =

Encryption formula

$C = M^e \bmod n$

$C = 5^7 \bmod 209$

$C = 78125 \bmod 209$

C = 168

Decryption formula :-

$M = c^d \bmod n$

$M = 168^{103} \bmod 209$

M = 6

5) Encryption technique :-

* It is the process of converting the plain text into cipher text

* It provides the user by providing security by hiding confidential information such as account number, password and transaction detail.

* The Encryption technique we are going to use are

  * Caesar cipher
  * vigenere cipher
  * Playfair cipher
  * AES-256-GCM
  * RSA - ASP

Ceaser cipher :-

* It uses the fixed number for the encryption process
* The keyword is given we want add to the plain text
* The formula for ceaser cipher text

$$c = (p+k) \bmod 26$$

* Eg : The plain text is : HELLO
* key : 3
* The ceaser cipher text is : KHOOR
* For the Decryption process use the formula $p = (c-k) \bmod 26$
* The plain text for the KHOOR is HELLO.

Playfair cipher :-

* It uses the five cross matrix
* It uses the one-column, one-Row and one-rectangular block code.
* It provide better security than ceaser cipher but it is vulnerable for cryat analysis

Vigenere cipher :-

* It uses the Repeating keyword for the encryption process.
* The keyword in the form of word.
* The formula for vigenere cipher text is

$$c = (p+k) \bmod 26$$

* Eg : The plain text P : HELLO
* Key : KEYE
* The ceaser cipher text is : RIJVS

the formula for Decryption is $P = (C - K) \bmod 26$

* The plain text for the Rijvs is HELLO

comparision table

Feature	Ceaser cipher	Playfair cipher	vigener cipher
Method	Single substitution	Paired substitution	Multiple substitution
Key	Number	keyword	Repeating key
security	very weak	weak	Better
Encryption	one letters	two letters	one letter

AES-256-GCM

* Uses a 256-bit key
* Provides confidentiality and integrity
* suitable for large financial data
* Detects unauthorized notification.

RSA:-

* uses public and private key
* helps securely exchange encryption keys
* support digital signatures by using supported digital schemas.

Modern Encryption Process:-

* Generate a random AES key.
* Encrypt customer information using AES-256-GCM
* Encrypt the AES key using the RSA public key
* send the Encrypted data and Encrypted key
* Recipient decrypt the AES key using the RSA private key.
* Decrypt the original message using the AES
* verify the authonication tag.

Conclusion:-

Classical encryption technique are used for education purpose only not for the modern financial ~~data~~ technique.

AES-256-GCM with RSA-OAEP key wrapping is the best choice.